

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

4.2 – Proposed Solution

The proposed system aims to analyze electricity consumption data to identify usage patterns, predict future demand, and provide actionable insights for efficient energy management. By leveraging data analytics, visualization tools, and machine learning techniques, the system helps consumers, utility providers, and policymakers make informed decisions regarding electricity usage.

Key Features of the Proposed Solution

1. Data Collection

- Gather electricity consumption data from smart meters, utility records, and public datasets.
- Collect historical and real-time usage information.

2. Data Storage

- Store collected data in a centralized database for easy access and management.
- Maintain historical records for trend analysis.

3. Data Preprocessing

- Clean and validate the collected data.
- Remove missing values, duplicates, and inconsistencies.

4. Consumption Pattern Analysis

- Analyze daily, weekly, monthly, and seasonal electricity usage trends.
- Identify peak and off-peak consumption periods.